MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE: INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT: COMPUTER SCIENCE AND ENGINEERING

PROGRAM: BACHELOR OF COMPUTER SCIENCE

COURSE NAME: APPLICATION DEVELOPMENT FOR MOBILE DEVICES

COURSE CODE: CS8306

LEVEL: THIRD YEAR

LECTURER NAME: MR. AMANI RUKOIJO

TASK NAME: GROUP ASSIGNEMENT

PROJECT NAME: DEVELOPMENT OF MUSTSO ELECTION SYSTEM

PARTICIPANTS NAME

S/NO.	NAME OF STUDENTS	REGISTRATION NO.	SIGNATURE
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GitHub Repository: https://github.com/dicksoneliezer/mustso-election

You Tube Link: https://youtu.be/j1GMPhwcm-8

INTRODUCTION

Mbeya University of Science and Technology (MUST) conducts regular student organization (MUSTSO) elections to elect student leaders who represent the student body in positions such as President, College MPs, and Department MPs.

Traditional voting uses paper ballots which causes long queues, human counting errors, delayed results, and high resource consumption.

Problem Statement

The current manual voting system faces several challenges:

• Inefficient vote casting

• Manual counting errors

• Delayed result announcement

• Difficulty verifying voter eligibility

• Lack of transparency

• High operational cost

PROJECT OBJECTIVES

Main Objective:

To develop a secure electronic voting system that improves efficiency, transparency, and accessibility.

Specific Objectives:

• Digital student registration

• Secure login system

• Candidate filtering by eligibility

• One vote per student per position

• Real-time results visualization

• Mobile responsive interface

• Administrative management tools

REQUIREMENTS ANALYSIS

Functional Requirements

Student Module:

- Register in the system
- Login securely
- View eligible candidates
- Cast vote
- View election results

Admin Module:

- Admin login
- Manage candidates
- View election results
- Monitor system statistics

Non-Functional Requirements

- Security
- Performance
- Availability
- Scalability
- Usability
- Responsiveness
- Data integrity

SYSTEM DESIGN & ARCHITECTURE

The system uses a 3-tier architecture:

Presentation Layer:

HTML, CSS, JavaScript

Application Layer:

PHP backend with authentication and voting logic

Database Layer:

MySQL database storing students, candidates, and votes

SECURITY & AUTHENTICATION

Security mechanisms implemented:

• Password hashing using bcrypt

• PHP session authentication

• SQL injection prevention using prepared statements

• XSS protection using htmlspecialchars()

• Access control for students and administrators

IMPLEMENTATION

Technology Stack:

Frontend:

HTML5, CSS3, JavaScript

Backend:

PHP

Database:

MySQL

Server:

Apache (XAMPP)

Version Control:

Git and GitHub

6. RESULTS & SCREENSHOTS

Include screenshots of:

- Home page
- Registration page
- Voting interface
- Results dashboard
- Admin dashboard
- Candidate management page
- Mobile responsive layout

7. CONCLUSION

The MUSTSO Election System successfully digitizes the voting process at Mbeya University of Science and Technology.

The system improves:

- Efficiency
- Security
- Transparency
- Accessibility

Students can vote electronically while administrators manage elections easily.

8. FUTURE IMPROVEMENTS

Future features may include:

- Email verification
- Two-factor authentication
- Mobile application version
- Blockchain-based voting
- Multi-language support